# Chapter 7. The Sieve

*You Shall Not Overcount*

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## 7.1 Enumerating the elements of intersecting sets

**Motivation.** Alternately subtracting and adding intersections corrects overcounts exactly, since $\sum_j (-1)^j \binom{s}{j} = 0$.

**Example 7.1**

14 students play soccer, 17 basketball, and 4 both. Then $14 + 17 - 4 = 27$ play at least one of the two.

**Example 7.2**

Add 18 hockey players; 4 play soccer and basketball, 3 soccer and hockey, 5 basketball and hockey, and 1 all three. Then $14+17+18-4-3-5+1 = 38$ play at least one (the all-three student is added, subtracted three times each, then added back).

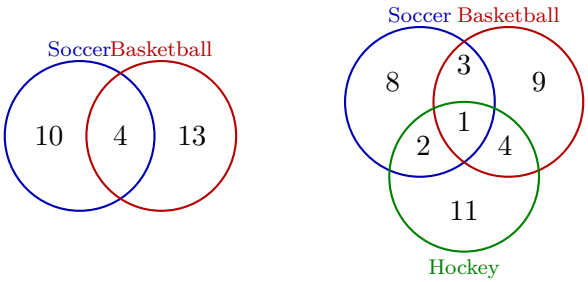



**Figure.** *Examples 7.1 and 7.2: region counts sum to* 27 *and* 38.

**Theorem 7.3 (Sieve formula, inclusion–exclusion)**

For finite sets $A_1, \dots, A_n$,

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \sum_{\{i_1,\dots,i_j\}} |A_{i_1} \cap \cdots \cap A_{i_j}|, \tag{7.1}$$

where $\{i_1, \dots, i_j\}$ runs over the $j$-element subsets of $[n]$.

**Proof.** For $x$ in the union let $S := \{i \mid x \in A_i\}$, $s := |S| \geq 1$; $x \in A_{i_1} \cap \cdots \cap A_{i_j}$ iff $\{i_1, \dots, i_j\} \subseteq S$, so $x$ is counted

$$s - \binom{s}{2} + \binom{s}{3} - \cdots + (-1)^{s-1} \binom{s}{s} = 1 - (1-1)^s = 1 \tag{7.2}$$

times; elements outside the union are never counted. ∎

## 7.2 Applications of the sieve formula

**Motivation.** The sieve counts objects with *none* of several bad properties.

**Example 7.4 (Derangements)**

How many permutations of $[n]$ have no fixed point (**derangements**; nobody gets their own hat)?

*Solution.* $A_i$: permutations fixing $i$. Each of the $\binom{n}{j}$ intersections of $j$ of them has $(n-j)!$ elements. By (7.1),

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^{n} (-1)^{j-1} \frac{n!}{j!},$$

so

$$D(n) = n! - |A_1 \cup \cdots \cup A_n| = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!}. \tag{7.3}$$

Hence $\frac{D(n)}{n!} \to e^{-1} \approx 37\%$. [Supplement] $D(n)$ is the integer nearest to $n!/e$ ($n \geq 1$): $\left|\frac{n!}{e} - D(n)\right| = n!\left|\sum_{i>n} \frac{(-1)^i}{i!}\right| < \frac{n!}{(n+1)!} \leq \frac{1}{2}$ (alternating, strictly decreasing terms).

**Theorem 7.5**

For positive integers $n, k$:

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n.$$

**Proof.** $k!\,S(n,k)$ counts surjections (Corollary 5.9). Let $A_i$ be the functions missing $i$; $j$-fold intersections have $(k-j)^n$ elements, so by the sieve there are $\sum_{j=1}^{k} (-1)^{j-1}\binom{k}{j}(k-j)^n$ non-surjections. Subtract from $k^n$. ∎

**Theorem 7.6 (Möbius inversion on subsets)**

Let $f, g$ be real functions on the subsets of $[n]$ with $g(S) = \sum_{T \subseteq S} f(T)$. Then

$$f(S) = \sum_{T \subseteq S} g(T)(-1)^{|S-T|}. \tag{7.4}$$

**Proof.** The right side is $\sum_{U \subseteq S} f(U) \sum_{U \subseteq T \subseteq S} (-1)^{|S-T|}$. $T$ is determined by $S \setminus T \subseteq S \setminus U$, so the inner sum is $\sum_i (-1)^i \binom{|S-U|}{i} = (1-1)^{|S-U|} = [U = S]$. ∎

## 7.3 Selected exercises

**Motivation.** Book exercises on the sieve and Möbius inversion.

**Exercise 7.2 (Three teams)**

A class of at least two students has three sports teams, and any two students are both members of at least one team. Prove that some team contains at least $\frac{2}{3}$ of the students.

**Solution.** $n \geq 2$ students, teams $T_1, T_2, T_3$.

*Step 1 (a student in only one team).* If $a$ is only in $T_1$, every $b$ shares a team with $a$, so $b \in T_1$: $T_1$ is everybody.

*Step 2 (no student in only one team).* Every student is in some team ($n \geq 2$), hence in at least two: $|T_1| + |T_2| + |T_3| \geq 2n$. ■

**Exercise 7.8 (Euler's function)**

Let $\varphi(n)$ be the number of integers in $[n]$ relatively prime to $n$. Prove that if $p_1, \ldots, p_k$ are the distinct prime divisors of $n$, then

$$\varphi(n) = n\Big(1 - \frac{1}{p_1}\Big) \cdots \Big(1 - \frac{1}{p_k}\Big).$$

**Solution.**

*Step 1 (the sets).* $A_i = \{x \in [n] : p_i \mid x\}$; $\varphi(n) = n - |A_1 \cup \cdots \cup A_k|$.

*Step 2 (intersections).* With $d_S = \prod_{i \in S} p_i$, $\bigcap_{i \in S} A_i =$ multiples of $d_S$ in $[n]$: $n/d_S$ elements.

*Step 3 (sieve).* By Theorem 7.3, with $d_\emptyset = 1$,

$$\varphi(n) = n - \sum_{\emptyset \neq S \subseteq [k]} (-1)^{|S|-1} \frac{n}{d_S} = \sum_{S \subseteq [k]} (-1)^{|S|} \frac{n}{d_S} = n \prod_{i=1}^{k} \Big(1 - \frac{1}{p_i}\Big),$$

expanding the product. ■

**Exercise 7.12 (The dual Möbius inversion)**

Let $h, r$ be real functions on the subsets of $[n]$ with $r(S) = \sum_{T \supseteq S} h(T)$ for all $S$. Prove that

$$h(S) = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

**Solution.** Complementation $X \mapsto \bar{X} = [n] \setminus X$ reverses inclusion.

*Step 1 (transport).* $f(A) = h(\bar{A})$, $g(A) = r(\bar{A})$; with $T = \bar{U}$,

$$g(A) = r(\bar{A}) = \sum_{T \supseteq \bar{A}} h(T) = \sum_{U \subseteq A} h(\bar{U}) = \sum_{U \subseteq A} f(U).$$

*Step 2 (apply Theorem 7.6).* $f(A) = \sum_{U \subseteq A} g(U)(-1)^{|A-U|}$.

*Step 3 (transport back).* $A = \bar{S}$, $U = \bar{T}$: $U \subseteq A \iff T \supseteq S$, $A - U = T - S$, so

$$h(S) = f(\bar{S}) = \sum_{T \supseteq S} g(\bar{T})(-1)^{|T-S|} = \sum_{T \supseteq S} r(T)(-1)^{|T-S|}.$$

■

**Exercise 7.23 (The derangement recurrence)**

Give a combinatorial proof of $D(n+1) = n(D(n) + D(n-1))$ for $n \geq 1$, without using (7.3). Here $D(0) = 1$ and $D(1) = 0$.

**Solution.** For a derangement $p$ of $[n+1]$, $k = p(n+1) \in [n]$: $n$ choices. For fixed $k$ we show $D(n-1) + D(n)$ derangements.

*Step 1 ($p(k) = n+1$).* $(k\ n{+}1)$ is a 2-cycle and the rest is any derangement of $[n] \setminus \{k\}$: $D(n-1)$.

*Step 2 ($p(k) \neq n+1$).* Delete $n+1$ from its cycle (length $\geq 3$): with $m = p^{-1}(n+1)$, $p'(m) = k$, $p' = p$ elsewhere on $[n]$. $p'$ is a derangement: $m \neq k$ since $p(k) \neq n+1$.

*Step 3 (the inverse of Step 2).* For a derangement $q$ of $[n]$ and $m = q^{-1}(k)$, set $p(m) = n+1$, $p(n+1) = k$, $p = q$ otherwise: a derangement with $p(k) = q(k) \neq n+1$, and the constructions are inverse. So $D(n)$.

Sum over $k$. (Check: $D(2) = 1 \cdot (0+1) = 1$.) ■

**Exercise 7.37 (No singleton blocks, by the sieve)**

Let $F(n)$ be the number of partitions of $[n]$ without singleton blocks. Express $F(n)$ through the Bell numbers.

**Solution.** Claim: $F(n) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j)$, $B(0) = 1$.

*Step 1 (the sets).* $A_i$: partitions with block $\{i\}$; $F(n) = B(n) - |A_1 \cup \cdots \cup A_n|$.

*Step 2 (intersections).* For $|J| = j$, $\bigcap_{i \in J} A_i$ = singletons of $J$ plus any partition of the other $n-j$: $B(n-j)$ elements, $\binom{n}{j}$ choices of $J$.

*Step 3 (sieve).* By Theorem 7.3,

$$F(n) = B(n) - \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} B(n-j) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} B(n-j).$$

E.g. $F(3) = 5 - 3 \cdot 2 + 3 \cdot 1 - 1 = 1$. Compare Exercise 5.34. ■